Ghaly Nicolas Jules

Software/Machine Learning Engineer
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Experience

Software Engineer at B-Line Express
Remote| March 2023 – August 2025

- Developed and maintained web-based applications using Vue.js, JavaScript, and Coldfusion.
- Implemented a Bayesian Lognormal distribution algorithm in Vue.JS to take data and create graphs for data analysis.
- Created reusable UI components, improving consistency across the product and reducing development time for new features.
- Implemented a template system for a mass email program tailored to upper management in non-profits.
- Collaborated closely with the UX/UI team to deliver engaging and responsive interfaces.

Software Engineer – Enterprise Data Management at Moody's Corporation
Remote | Summer 2022

- Worked in a Scrum environment, closely with designers and backend developers to meet client requirements.
- Query and analyze data housed within Athena AWS to create and design dashboards for QA and data analysis.
- Crafted detailed overview and process flow of complex Python scripts for business users to use as a reference guide.
- Worked at the intersection of data, sustainability and technology to support an ESG Data product using technologies such as AWS and Python programming language.

Web and Mobile Developer – at GAMMA Research Group
Remote | January 2021 – May 2022

- Led the development of frontend functionalities using React.JS, allowing users to interact with data coming from connected devices, monitor different data measurements, and export a sorted and organized dataset into a self-generated spreadsheet.
- Query and analyze data housed within Athena AWS to create and design dashboards for QA and data analysis.
- contributed to the integration of RESTful API services to connect the application backend to a React.JS based frontend.
- Assisted with the development of mobile applications to monitor and regulate the devices using React Native connecting to the backend.

Skills

- **Programming / Languages**: JavaScript, TypeScript, Python, HTML/CSS, SQL
- **Frameworks**: React, Vue.js, Tailwind CSS, Node.js, Next.js, Django, Streamlit
- **ML / Data Science**: scikit-learn, pandas, NumPy, Matplotlib, Seaborn, Logistic Regression, Random Forest, KNN, cross-validation, hyperparameter tuning
- **Tools**: Git, AWS, VS Code, Jupyter Notebook, Anaconda

Projects

Heart Disease Risk Predictor | GitHub | Live Demo
An end-to-end machine learning application that predicts heart disease risk from clinical patient data, deployed as an interactive web application.

- Built and compared three classification models (Logistic Regression, KNN, Random Forest) using scikit-learn, achieving **~85% accuracy** and **~91% recall** via 5-fold cross-validation.
- Performed hyperparameter tuning with RandomizedSearchCV and GridSearchCV, and conducted full model evaluation including ROC/AUC, confusion matrix, precision, recall, and F1-score.
- Analyzed feature importance via logistic regression coefficients to derive clinical insights from model predictions.
- Deployed as a Streamlit web application with a real-time prediction interface accessible via public URL.

A-Build | Github
A task tracking web app built to help users organize their daily activities efficiently, track their steps, bugs and progress throughout the development of a project.

- Developed using React.js and Tailwind CSS for a responsive and clean user experience.
- Integrated Redux for state management and Axios to handle API requests.
- Utilized Django in the backend with Django Rest Framework for RESTful API services and AllAuth for token authentication.

Census | Github
A mobile application designed to streamline the acquisition of population census data in Haiti.

- Built with Java, MySQL, and Android Studio providing a robust, full stack application.
- Utilized PHP to build RESTful APIs and Axios to handle those API requests
- Integrated Shared Preferences for state management.

Education

Master of Science in Computer Engineering
Georgia Institute of Technology | 2021 – 2022

Bachelor of Science in Computer Science
ESIH | 2015 – 2019

Languages

- **English**: Fluent
- **French**: Fluent
- **Haitian Creole**: Fluent
- **Spanish**: Basic

Leadership Activities

B.E.L Initiative Accelerator (Georgia Haitian American – Chamber of Commerce)
Fellow and Co-Founder of Haiti RELAY

- Co-led the development and launch of Haiti RELAY, a venture with the Georgia Institute of Technology.
- Collaborated in the development of a multi-phase business plan, which outlines the production and sales of solar panel systems, as panel efficiency improves over time and manufacturing becomes more cost-efficient.
- Manufactured and sold 25 low-cost and portable solar panel systems, capable of providing light and cell phone power.

Compassion International Haiti, Project HA-254
Program Advisor

- Organized and led a series of 5 workshops for 150 Haitian high-school students to provide advice and guidance in career selection and professional development.
- Facilitated mentoring sessions to help students define and execute community impact projects.